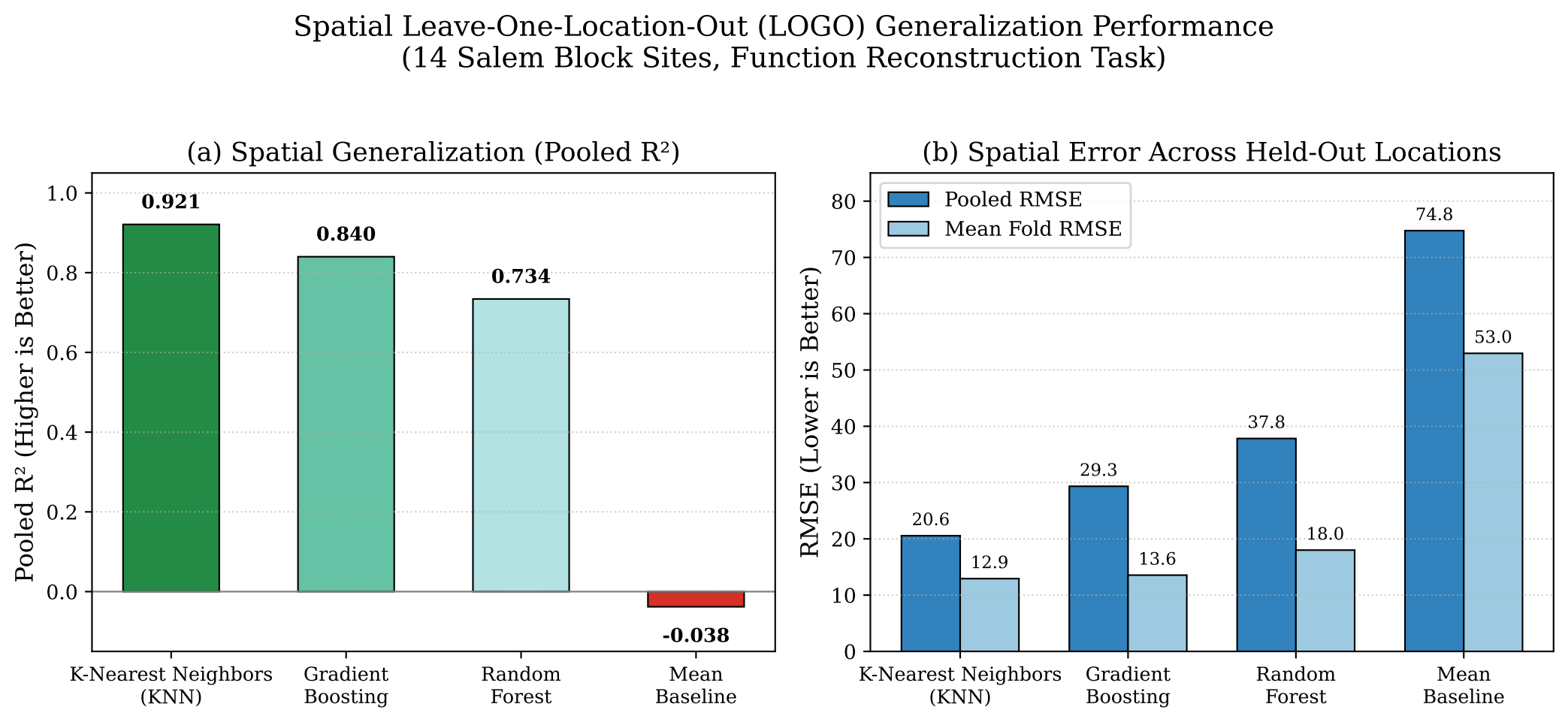
Spatial Leave-One-Location-Out (LOGO) Generalization Performance
(14 Salem Block Sites, Function Reconstruction Task)
(a) Spatial Generalization (Pooled R²)
Pooled R² (Higher is Better)
1.0
0.8
0.6
0.4
0.2
0.0
0.921
0.840
0.734
-0.038
K-Nearest Neighbors (KNN)
Gradient Boosting
Random Forest
Mean Baseline
(b) Spatial Error Across Held-Out Locations
RMSE (Lower is Better)
Pooled RMSE
Mean Fold RMSE
80
70
60
50
40
30
20
10
0
20.6
12.9
29.3
13.6
37.8
18.0
74.8
53.0
K-Nearest Neighbors (KNN)
Gradient Boosting
Random Forest
Mean Baseline